ML System Deployment
Module

HYBRID RISK-AWARE HEALTH INSURANCE RECOMMENDATION SYSTEM

SEMANTIC RETRIEVAL	RULE-BASED FILTERING	FAISS + BM25 LLAMA 3.3 70B

TEAM FANTASTIC 4

INTRODUCTION

Why finding the right health insurance is hard?

Health insurance policies are:

- Complex and lengthy
- Difficult to compare
- Not tailored to individual medical needs

Manual PDF Review

Hundreds of pages of policy text per plan

Coverage Blind Spots

Exclusions, waiting periods, condition-specific limits

No Condition Matching

Cannot handle pre-existing condition requirements

Zero Interpretability

No explanation of why a plan was recommended

Irish Health Insurance Market

19	Active Plans Analysed
950	Policy Text Chunks
4	Major Providers
2.4s~	Total Query Latency

OUR APPROACH

A two-layer AI pipeline for risk-aware recommendations

FAISS

HYBRID RETRIEVAL

FAISS + BM25 + RRF

Semantic + keyword search

Retrieves top 10 candidate plans

Combines results using RRF

950 policy text chunks

RULE ENGINE

Deterministic Scoring

Hard filters (Serious Issues)

Age validation (soft check)

Utilisation-based scoring

Condition coverage proxy

LLM RERANKER

Llama 3.3 70B via Groq

Evaluates shortlisted plans

Uses policy text + numeric data

Considers medical conditions, usage

RECOMMENDED

OUTPUT

Top 3 Ranked Plans

Fit score + rule score

Strengths & weaknesses

Coverage gaps

Source policy link

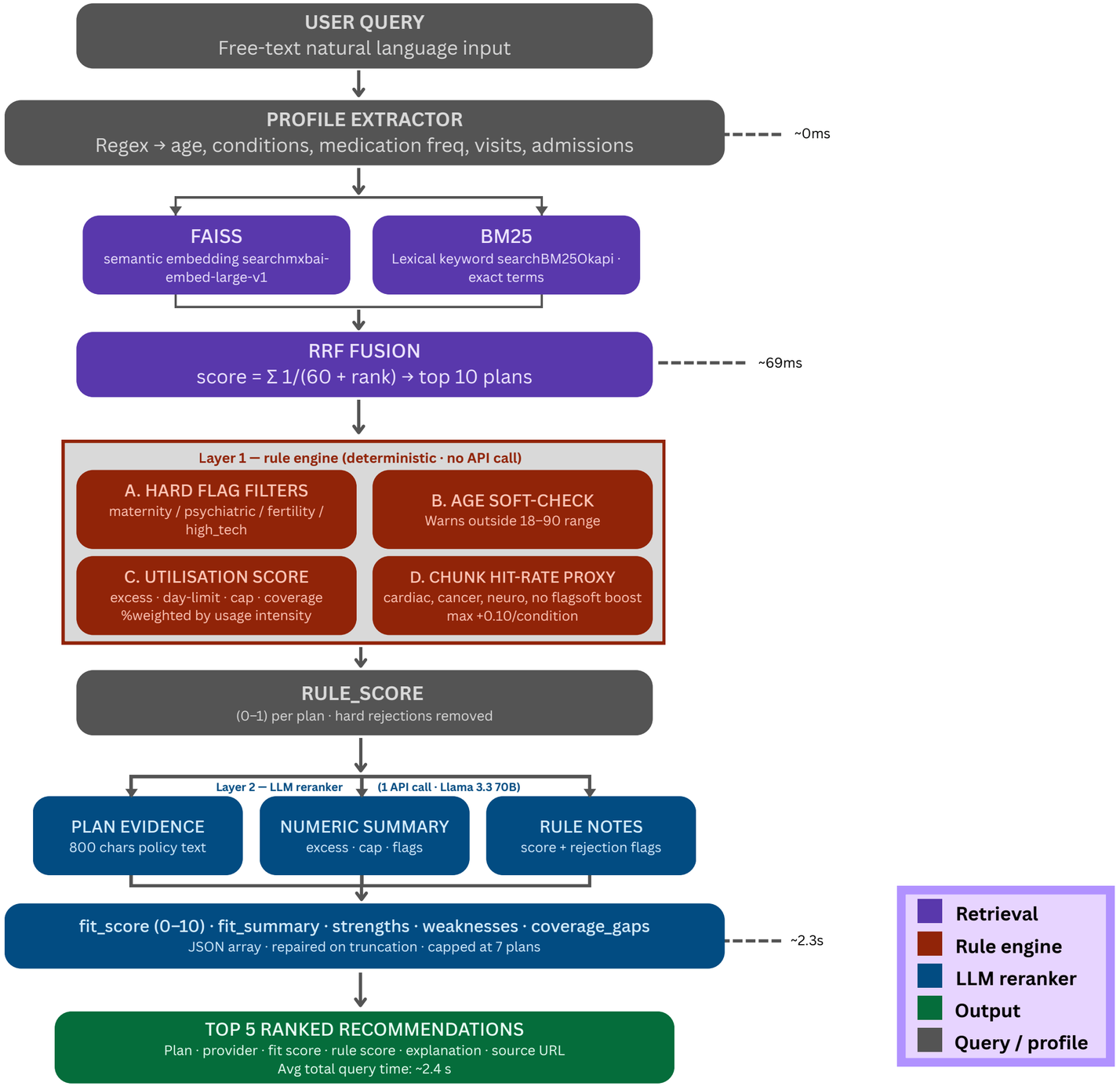
Annotated PDF

WORKFLOW

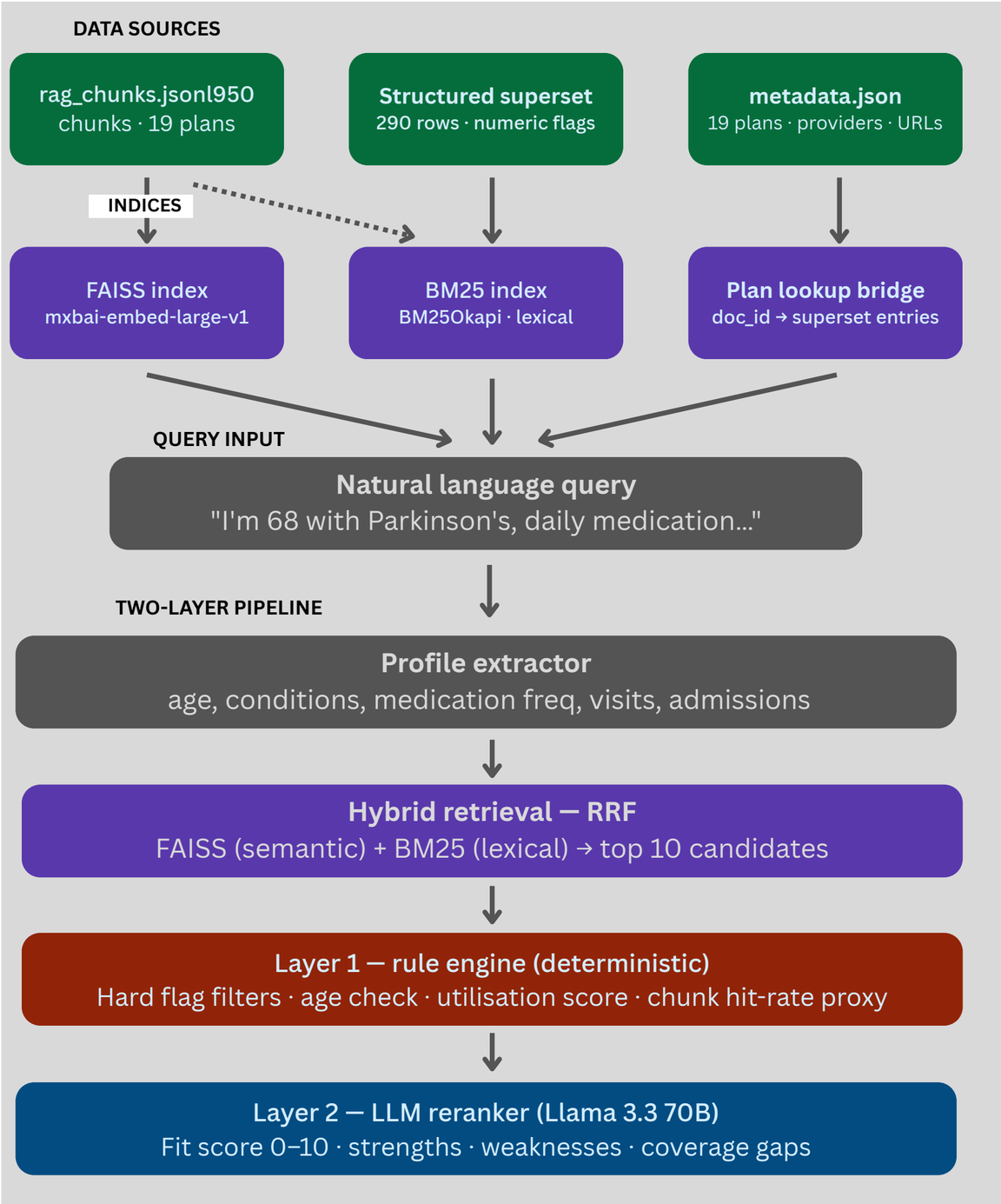
Hybrid Retrieval

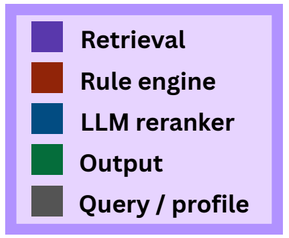
Deterministic Scoring

LLM Rank and Explain



ARCHITECTURE





DATA ARCHITECTURE

Multi-source pipeline across 4 Irish health insurers

rag_chunks.jsonl

950 chunks

19 plans · policy text

Structured Superset

290 rows

Numeric flags & limits

metadata.json

19 plans

Providers · source URLs

COVERAGE BY PROVIDER

Irish Life Health	plans 7
Laya Healthcare	plans 4
Level Health	plans 5
Vhi	plans 3

STRUCTURED SUPERSET FIELDS

excess_amount	copayment_amount
coverage_percentage	limit_amount
day_limit	limit_scope
maternity_flag	psychiatric_flag
fertility_flag	high_tech / international_flag

PROJECT TIMELINE

PHASE 1

FAISS (MiniLM) + BM25

Rule-based scoring

Weak semantics

Chunk fragmentation

Missing Rule Scores

PHASE 2

MXBAI embeddings

Better semantic matching

Removed BM25: lost lexical precision

Poor handling of numeric values

Poor Scenario

No reasoning layer

PHASE 3

Hybrid
(FAISS + BM25 + RRF)

Rule engine + LLM

PDF annotator

Balanced precision + semantics

Explainable results

HYBRID RRF RETRIEVAL

FAISS semantic search + BM25 lexical search, fused via Reciprocal Rank Fusion

FAISS Semantic Search

- Dense embeddings of policy text
- Understands meaning (not keywords)
- Finds related plans (e.g., “oncology” ≈ “cancer”)
- Based on cosine similarity

BM25 Lexical Search

- Exact keyword matching
- Strong for medical terms & numbers
- Handles terms like “maternity”, “€ excess”
- No embeddings (statistical scoring)

RRF FUSION FORMULA: Combines both rankings, Rewards agreement between methods

DETERMINISTIC RULE ENGINE

Hard correctness guarantees before the LLM ever sees a plan

A HARD FLAG FILTERS

NO API CALL

- Removes ineligible plans
- Checks: maternity, psychiatric, fertility, high-tech
- Uses structured data only

B AGE SOFT-CHECK

WARN ONLY

- Valid age range: 18–90
- Outside range → warning only
- No plan rejection

C UTILISATION SCORE

NUMERIC

- Inputs Medication frequency, Specialist visits, Hospital admissions
- Evaluates Excess, Day limits, Coverage %, Benefit caps

D CHUNK HIT-RATE PROXY

RAG PROXY

- Handles missing structured conditions
- Uses keyword presence in RAG chunks
- Boost: up to +0.10 per condition

Rule Score (0–1) per plan | Hard filtering applied | ~7–10 plans passed to LLM

LLM RERANKER

Llama 3.3 70B (Groq) · Context-aware reasoning

LLM INPUT (PER PLAN, ≤7 PLANS)

User profile (age, conditions, utilisation)

Up to 800 chars of actual policy text

Numeric summary (excess, cap, day-limit, flags)

Rule engine score and rejection flags

LLM OUTPUT (JSON ARRAY)

Fit Score (0–10)

Plain-English Summary

Strengths (Green signals)

Weaknesses (Red signals)

Coverage Gaps / Risk Conditions

ADVANCED IMPLEMENTATION

Complex RAG optimisation + hybrid search/reranking

Hybrid RRF Search

- FAISS + BM25
- Reciprocal Rank Fusion

Two-Stage Reranking

- Rule engine → correctness
- LLM → nuanced reasoning

Superset Bridge Lookup

- doc_id → plan names
- Handles inconsistent formats

JSON Repair + Token Safety

- JSON repair
- Token-safe input (~7 plans)

Utilisation-Weighted Scoring

- Penalises high excess
- Based on usage patterns

Chunk Hit-Rate Proxy

- Handles missing conditions
- Keyword-based boost

STREAMLIT APPLICATION

3 Plans Side-by-Side

Colour-coded fit, rule and RRF scores in responsive columns

Inline PDF Viewer

Policy PDFs with highlighted evidence chunks

Strengths / Weaknesses

Green / red / amber indicators for quick understanding

Coverage Gap Detection

Identifies weak coverage areas from LLM reasoning

PDF Cache & Download

Store and download highlighted PDFs via sidebar

SAMPLE OUTPUT

Query: "I'm 68 with Parkinson's. Neurologist monthly, daily medication."

RANK 1 • Plan D (Table of Cover) • Level Health

Fit Score	Rule Score	RRF Score
8 / 10	0.895	Top 10

Download annotated pdf

This plan suits the user due to its low excess and high day limits, beneficial for a Parkinson's patient requiring frequent medical visits and daily medication. The LLM identifies strong outpatient and specialist cover from the policy text, even though no explicit neurological flag exists in the structured data.

STRENGTHS

- Low excess of €38
- 100 day hospital cover
- Strong outpatient allowance

WEAKNESSES

- No explicit neuro flag in data
- High copayment €1,000
- Private hospital list limited

COVERAGE GAPS

- Neurological conditions
- Diabetes (no flag)
- High-tech drug uncertain

EVALUATION METRICS

Retrieval Quality	Context Precision	4.62 / 5
	Context Recall	4.34 / 5
LLM Quality	Factual Consistency	4.64 / 5
	Instruction Following	5.0 / 5
Latency	Retrieval Time	0.07 s
	LLM Latency	2.33 s
	Total Query Time	2.40 s
Generation Performance	TTFT	0.33 s
	TPOT	0.0021 s/token
	Throughput	469 tok/s
	Output Tokens	~939

Thank You

TEAM FANTASTIC 4

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github.com/AshwinPrasanth/ML-Deployment